23R–D25 (PHY5304C)

2025

PHYSICS

(Major–III)

Paper : PHY5304C

(Atomic and Nuclear Physics)

Full Marks : 45

Time : Two hours

The figures in the margin indicate full marks for the questions.

1. Answer the following questions : $1 \times 4 = 4$

(a) What is an endoenergic nuclear reaction ?

(b) Can a neutrino be accelerated by an accelerator ? Give justification.

(c) Write the electronic configuration of $Cr(z = 24)$.

(d) State Aufbau principle.

2. Answer the following questions : $2\times3=6$

(a) Calculate the Q-value of the nuclear reaction ${}_1H^3 + {}_1D^2 \rightarrow {}_2He^4 + {}_0n^1$ where the exact mass of ${}_1H^3$ is 3.0160 *amu*, ${}_1D^2$ is 2.0141 *amu* and ${}_2He^4$ is 4.0026 *amu*.

(b) Calculate spin and parity of the nucleus Na^{23}.

(c) Calculate the magnitude of orbital magnetic moment d-electron.

$\mu_e = \frac{e}{2m} L$

3. Answer ***any three*** the following questions: $5\times3=15$

(a) Establish the Geiger–Nuttal law.

(b) Establish the Weisacker's semi empirical mass formula.

OR

(c) Establish the conditions for $+\beta$ and $-\beta$ -decay. $2\frac{1}{2}+2\frac{1}{2}=5$

(d) State Larmor theorem and deduce an expression for Larmor frequency. $1+4=5$

$\tau = \mu_l \times B$

OR

(e) Explain LS coupling with its selection rules.

4. Answer (a) **or** (b) from the following questions :

(a) Explain the construction and working principle of a GM counter with a neat diagram. 10

OR

(b) Explain the construction and theory of a Cyclotron with a neat diagram. 10

5. Answer (c) **or** (d) from the following questions :

(c) Deduce the Rutherford's α-scattering formula. 10

OR

(d) State the postulates of Bohr's atomic model. Deduce an expression for energy of an electron in the n-th Bohr's orbit. Find the wavelength of the second line of hydrogen Balmer series.

2+6+2=10

E= avR